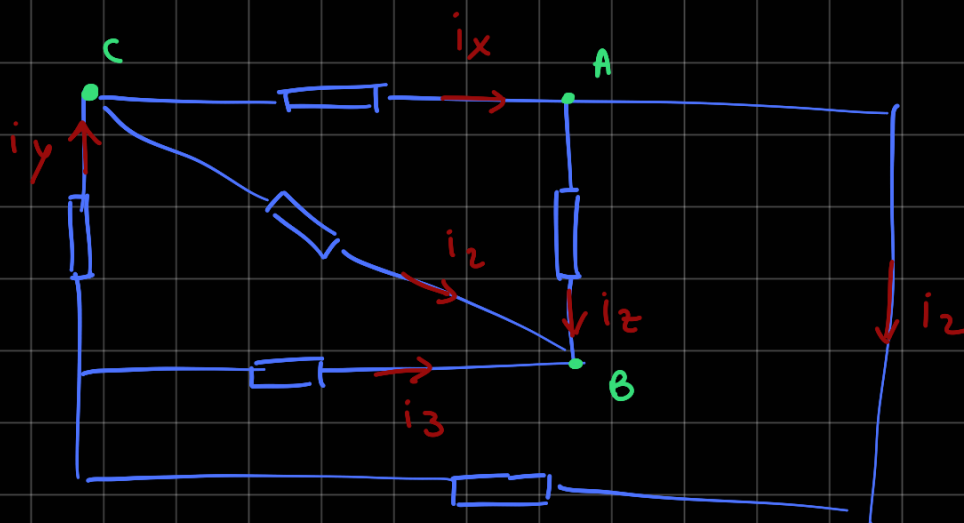


1



Trovo i_z con LKC sul nodo A

$$i_z + i_1 - i_3 = 0 \Rightarrow i_z = i_3 - i_1 = 2A$$

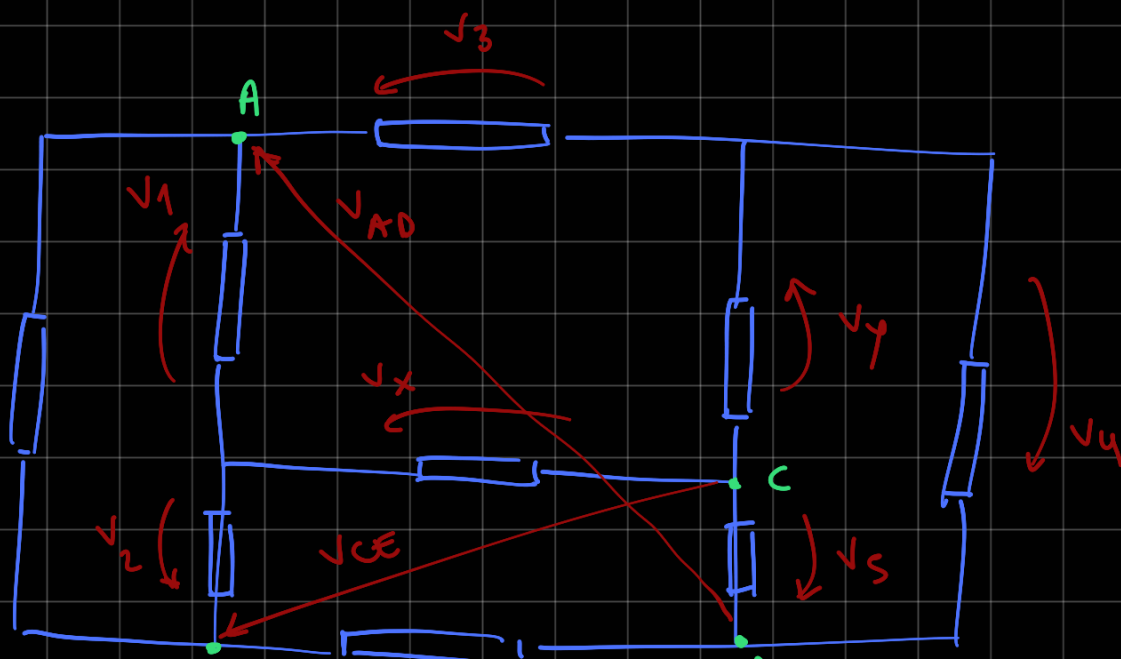
Trovo i_x con LKC sul nodo B

$$i_x + i_2 - i_z = 0 \Rightarrow i_x = i_z - i_2 = 4A$$

Analogamente calcolo i_y con LKC su C

$$i_y - i_1 - i_x = 0 \Rightarrow i_y = i_x + i_1 = 5A$$

2



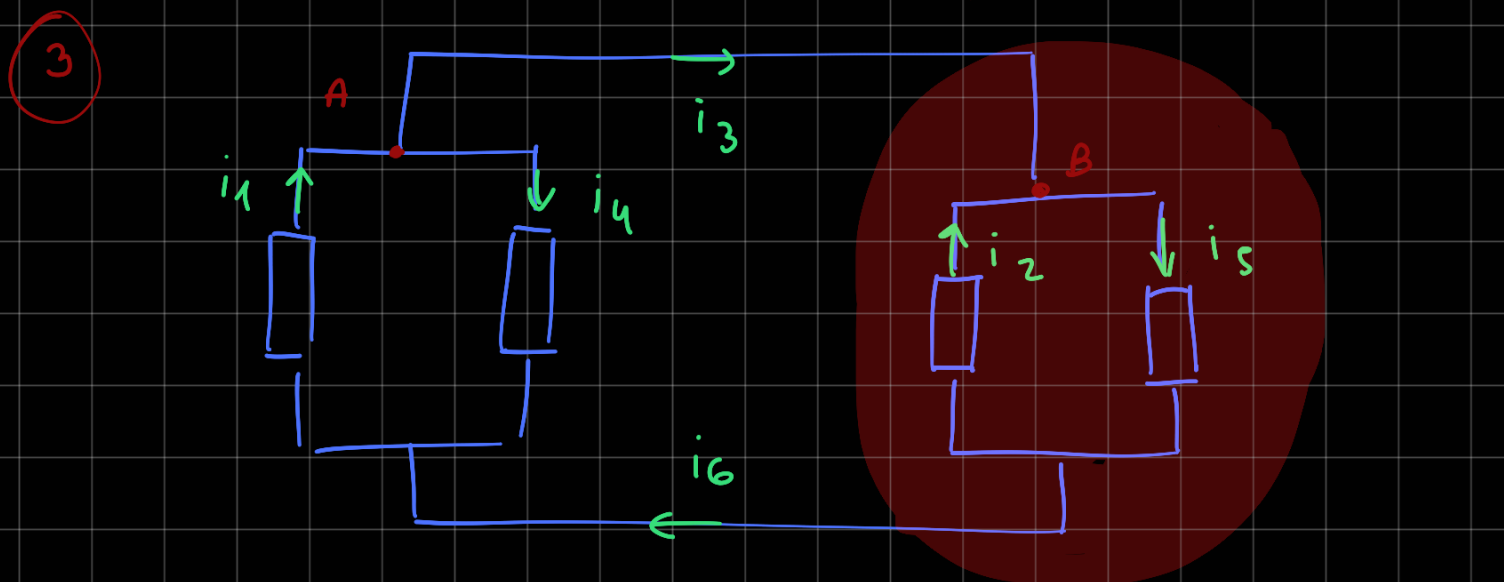
$$V_y + V_4 - V_5 = 0 \Rightarrow V_y = V_5 - V_4 = -4V$$

$$V_x + V_1 - V_3 - V_y = 0 \Rightarrow V_x = V_y + V_3 - V_1 = -6V$$

$$V_z - V_2 - V_x + V_5 = 0 \Rightarrow V_z = V_x + V_2 - V_5 = 0V$$

$$-V_{AD} + V_z - V_2 + V_1 = 0 \Rightarrow V_{AD} = V_z + V_1 - V_2 = -1V$$

$$-V_{CE} - V_z - V_5 = 0 \Rightarrow V_{CE} = -V_z - V_5 = 2V$$



$$\text{LKC}(A): i_1 - i_4 - i_3 = 0 \Rightarrow$$

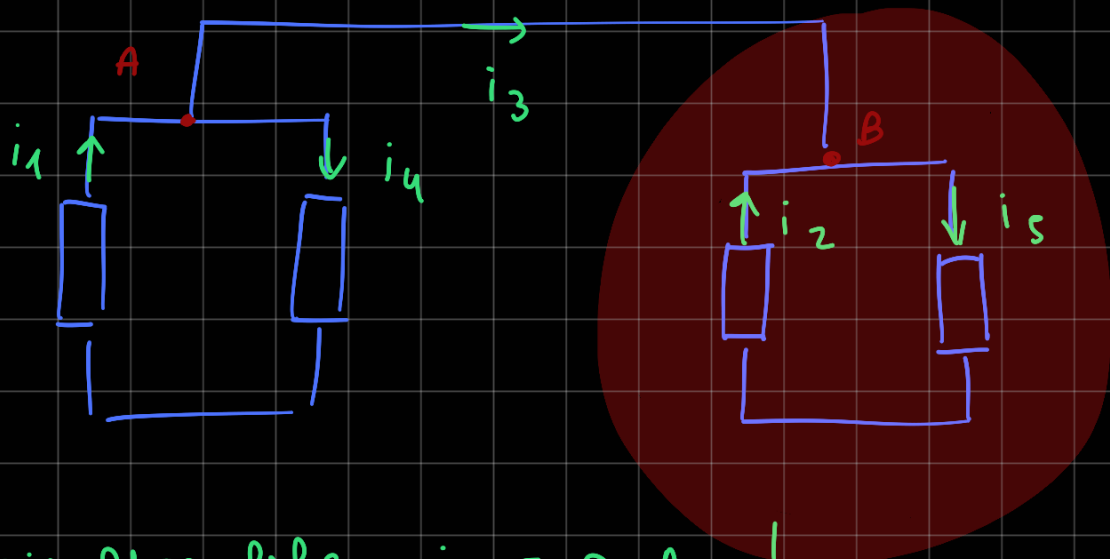
$$\Rightarrow i_1 = i_4 + i_3 = 7A$$

$$\text{LKC}(B): i_2 + i_3 - i_5 = 0 \Rightarrow$$

$$\Rightarrow i_2 = i_5 - i_3 = 0A$$

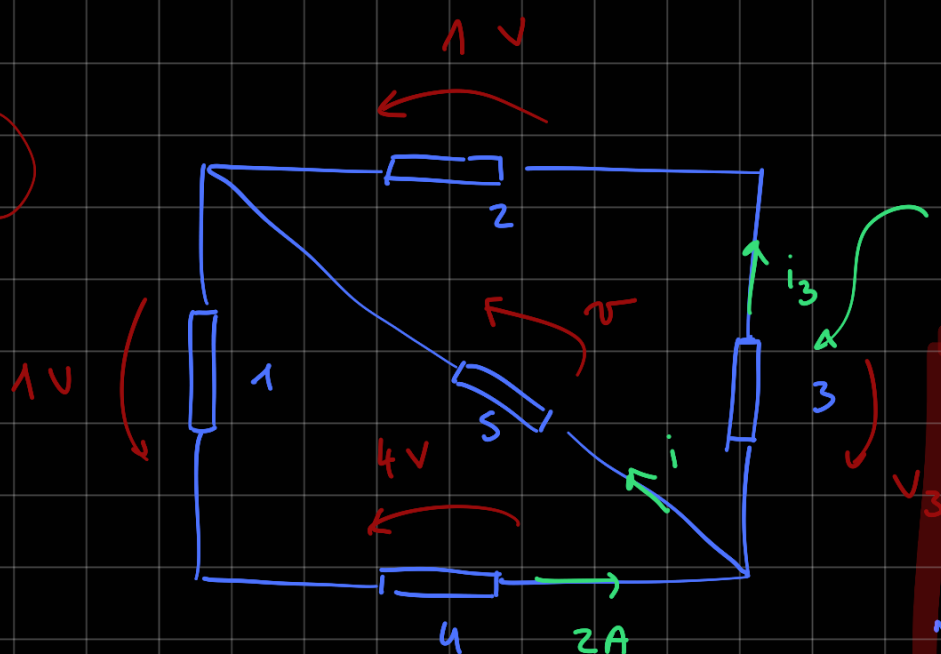
$$\text{LKC}(C): i_6 - i_3 = 0 \Rightarrow i_6 = i_3 = 3A$$

OSSERVA: se forse stato così...



risulterebbe $i_3 = 0A$!

4



20 W
"assorbita"
(quindi è la
convenzione
degli
utilizzatori, e
metterò i_3 e V_3
con il verso opportuno)

$$\text{LKT}(1-4-3): v + 1V - 4V = 0 \Rightarrow v = 3V$$

$$\text{LKT}(3-2-3): v - 1V + V_3 = 0 \Rightarrow$$

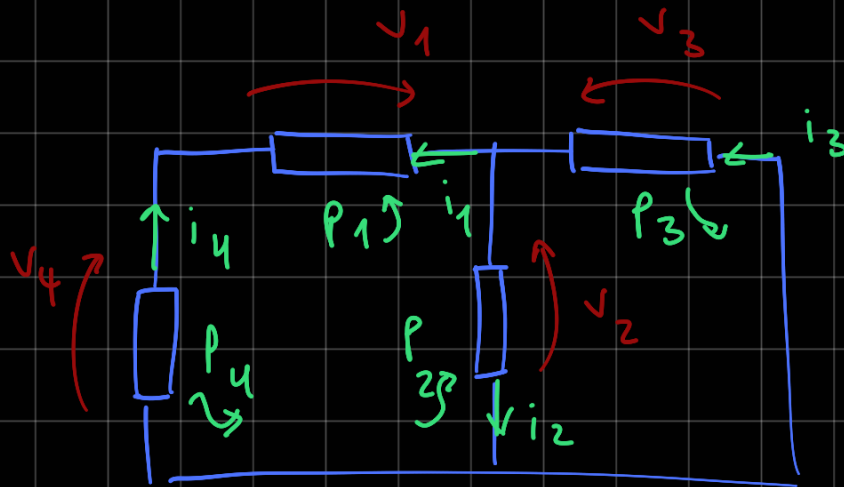
$$\Rightarrow V_3 = 1V - v = -2V$$

$$P_3 = V_3 \cdot i_3 \Rightarrow i_3 = \frac{P_3}{V_3} = -10A$$

$$\text{LKC (NODO 4-5-3)}: i + i_3 - 2A = 0 \rightarrow$$

$$\Rightarrow i = 2A - i_3 = 12A$$

5



$$p_4 = v_4 \cdot i_4 \Rightarrow i_4 = \frac{p_4}{v_4} = 2A$$

$$\Rightarrow i_1 = -i_4 = -2A$$

$$\Rightarrow p_1 = v_1 \cdot i_1 = -6W$$

$$\text{LKT (MAGLIA 1-2-4)}: v_2 = v_1 + v_4 = 8V$$

$$\text{LKC (NODO 1-2-3)}: i_2 = i_3 - i_1 = 3A$$

$$\Rightarrow p_2 = i_2 v_2 = 24W$$

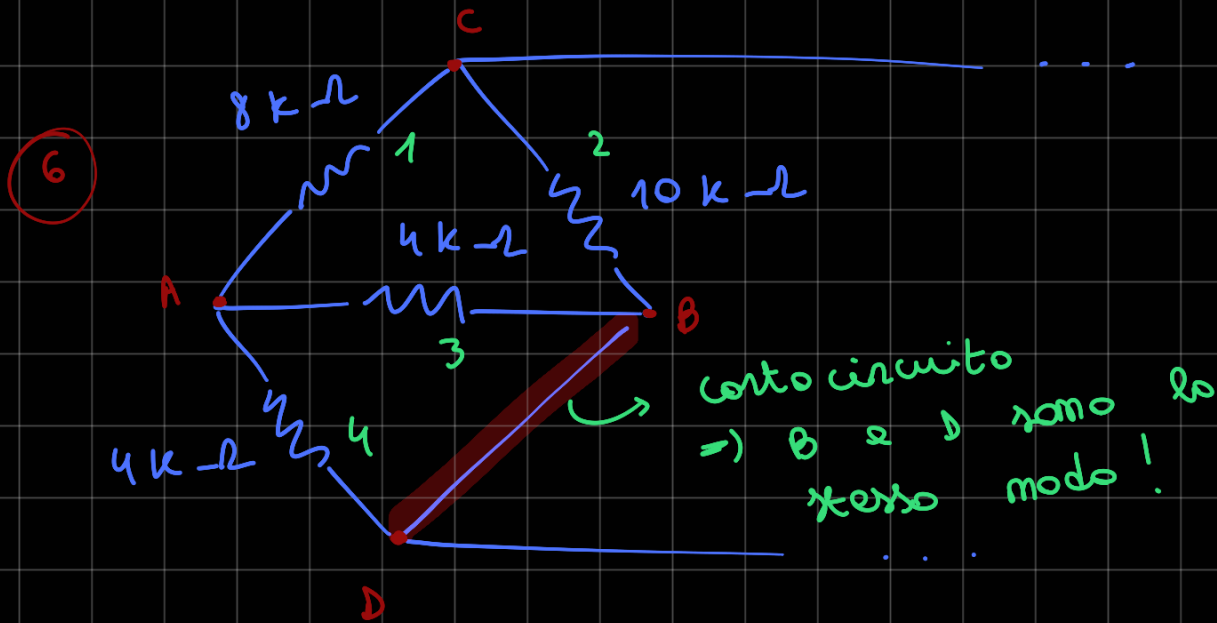
$$\text{LKT (MAGLIA 2-3)}: v_3 = v_2 = 8V$$

$$\Rightarrow p_3 = i_3 v_3 = 8W$$

verifico il teorema di Tellegen...

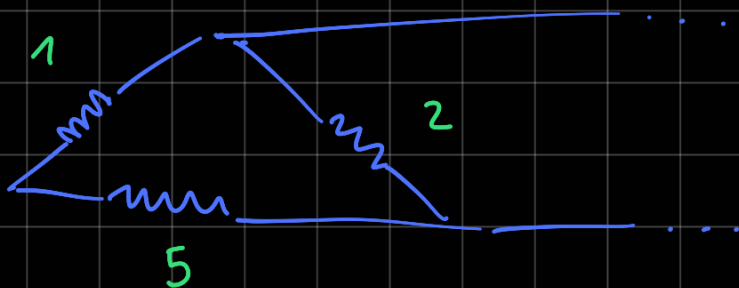
GENERATORI: $p_4 + p_3 = 18 \text{ W}$

UTILIZZATORI: $p_2 + p_3 = 18 \text{ W}$



3 e 4 sono in parallelo!

circuito equivalente...

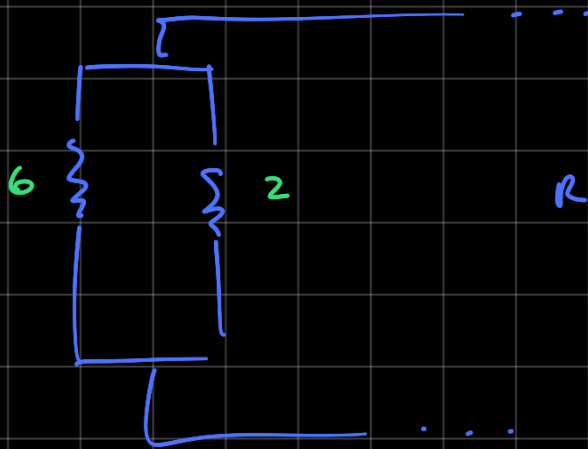


$$\frac{1}{R_5} = \frac{1}{R_3} + \frac{1}{R_4} \Rightarrow R_5 = \left(\frac{1}{R_3} + \frac{1}{R_4} \right)^{-1} = 2 \text{ k}\Omega$$

ora, 1 e 5 sono in serie!

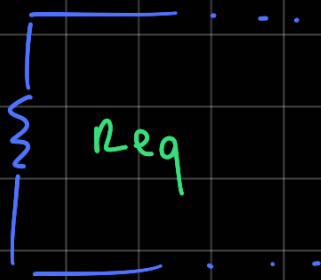
circuito eq. 2:

circuito eq. 2.



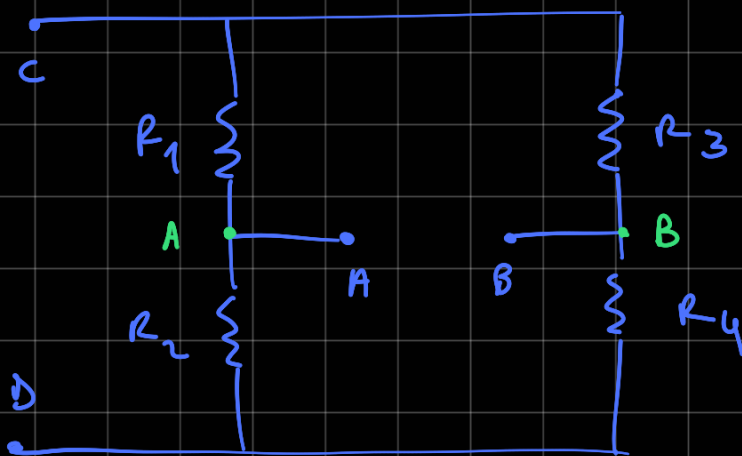
$$R_6 = 2k + 8k = 10k \Omega$$

Infine, 6 e 2 sono in parallelo!



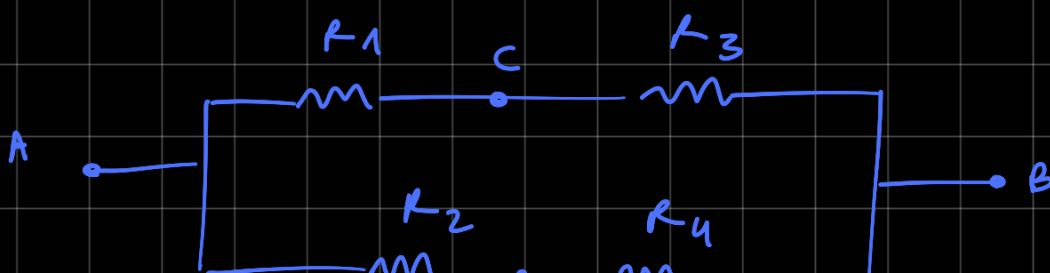
$$R_{eq} = \left(\frac{1}{R_5} + \frac{1}{R_2} \right)^{-1} = 5k \Omega$$

7



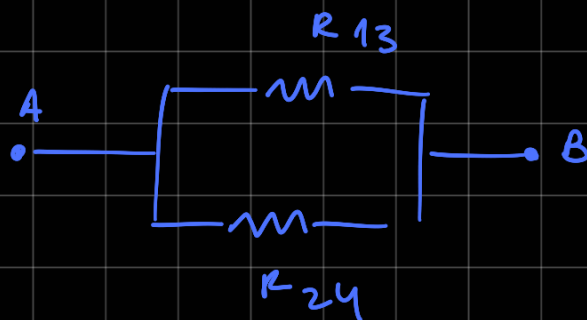
1° CASO: cd e' aperto

Il circuito e' equivalente a...



R_1 e R_3 sono in serie, e analogo con R_2 e R_4

circuito eq. :

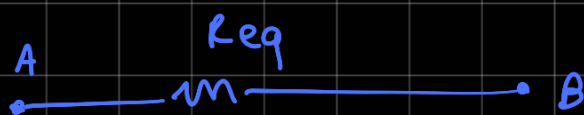


$$R_{13} = R_1 + R_3 = 25 \, \Omega$$

$$R_{24} = R_2 + R_4 = 25 \, \Omega$$

R_{13} e' in parallelo a R_{24}

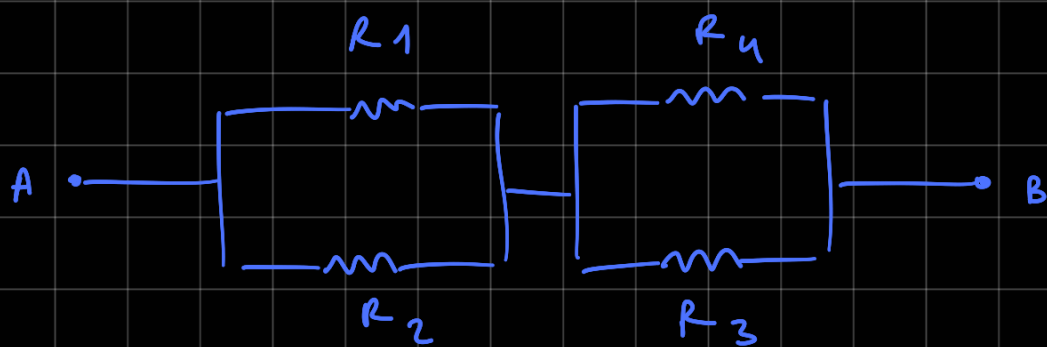
circuito eq. 2 :



$$R_{eq} = \left(\frac{1}{R_{13}} + \frac{1}{R_{24}} \right)^{-1} = 12,5 \, \Omega$$

2° CASO: cd e' cortocircuitato

il circuito e' equivalente a...



⇒ Il circuito equivalente è

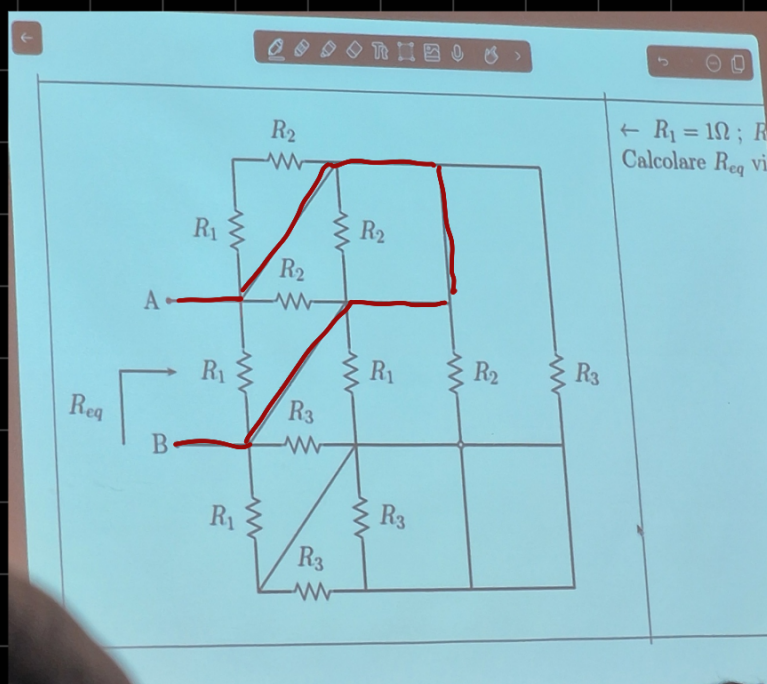


$$R_{12} = \dots = \frac{20}{3} \Omega$$

$$R_{34} = \dots = \frac{15}{4} \Omega$$

$$\Rightarrow R_{eq} = R_{12} + R_{34} = \frac{125}{12} \Omega$$

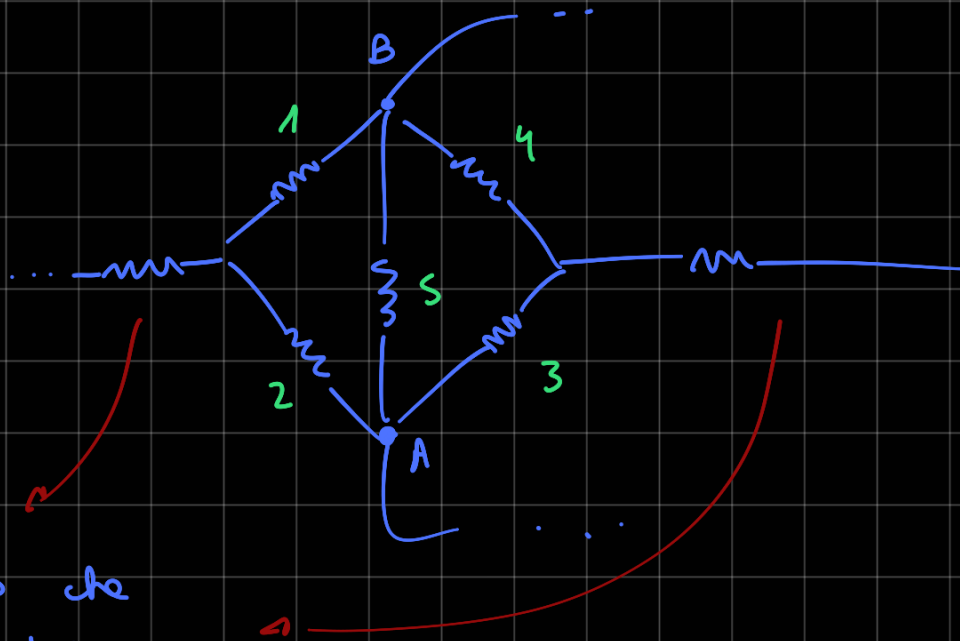
8



poiché A e B sono connessi tramite un corto circuito, significa che i nodi A e B coincidono

$\Rightarrow$ la resistenza equivalente in A e B è 0Ω

9

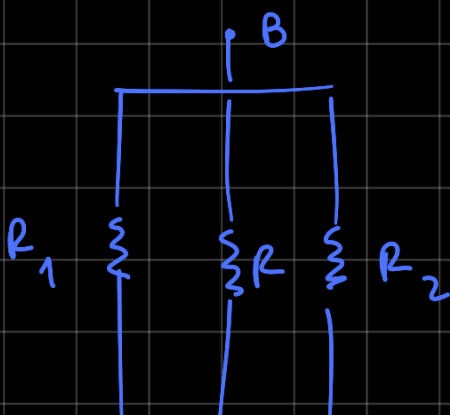


Notiamo che queste due resistenze sono zero (puoi prendere un modo generalizzato che comprende tutto tranne una delle due resistenze)

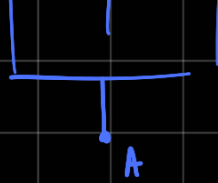
Notiamo che 1 e 2 sono in serie, e analogo per 3 e 4

poniamo dirlo solo perché abbiamo dimostrato che è come se non ci fossero i due resistori ai lati!

circuito eq.:

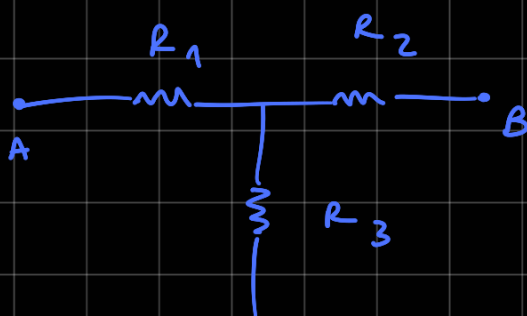


$$R_1 = R_2 = R + R = 2R$$



poiche' R_1 , R e R_2 sono in parallelo...

$$R_{eq} = \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R} \right)^{-1} = \frac{R}{2}$$



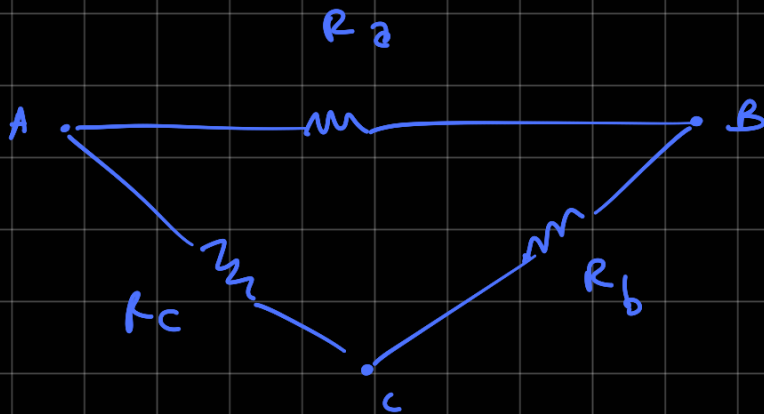
$$R_1 = 10 \, \Omega$$

$$R_2 = 20 \, \Omega$$

$$R_3 = 40 \, \Omega$$

10

Si può trasformare una configurazione
a STELLA in una a TRIANGOLO



Per farlo, le formule sono...

$$R_a = \frac{R_1 R_2 + R_2 R_3 + R_1 R_3}{R_3} \longrightarrow \text{opposta nella stella}$$

$$R_b = \frac{R_1 R_2 + R_2 R_3 + R_1 R_3}{R_1}$$

$$R_c = \frac{R_1 R_2 + R_2 R_3 + R_1 R_3}{R_2}$$

In questo caso...

$$R_a = 35 \, \Omega$$

$$R_b = 140 \, \Omega$$

$$R_c = 70 \, \Omega$$

OSSERVA: se vuoi passare da triangolo a stella...

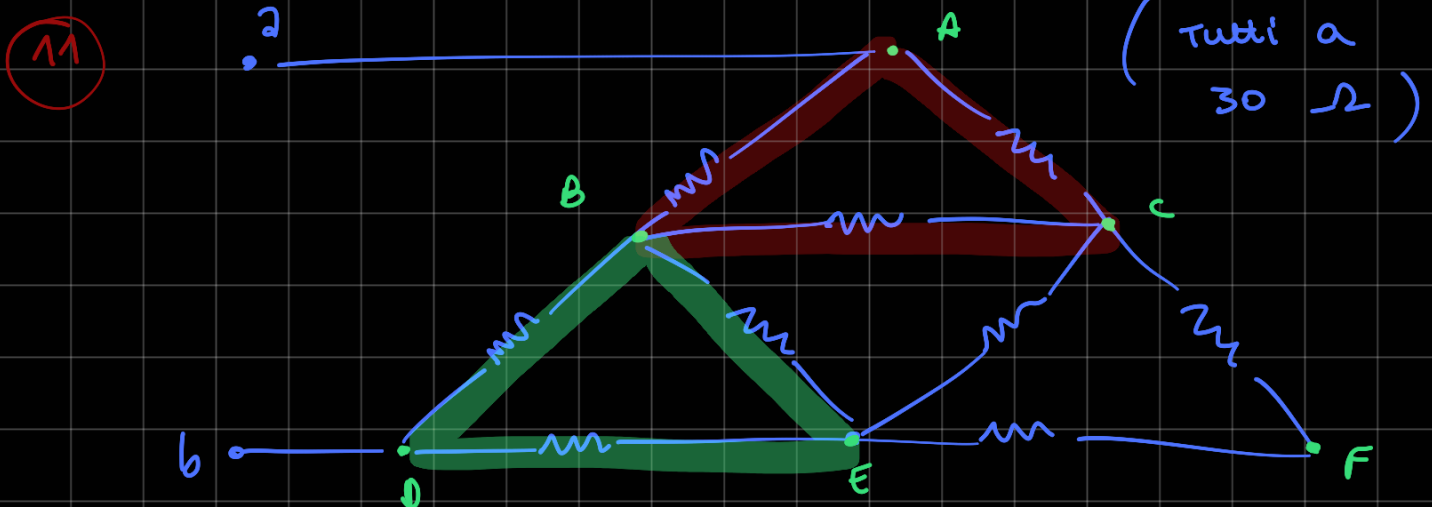
$$R_1 = \frac{R_a R_c}{R_a + R_b + R_c} \rightarrow \text{I due resistori vicini}$$

$$R_2 = \frac{R_a R_b}{R_a + R_b + R_c}$$

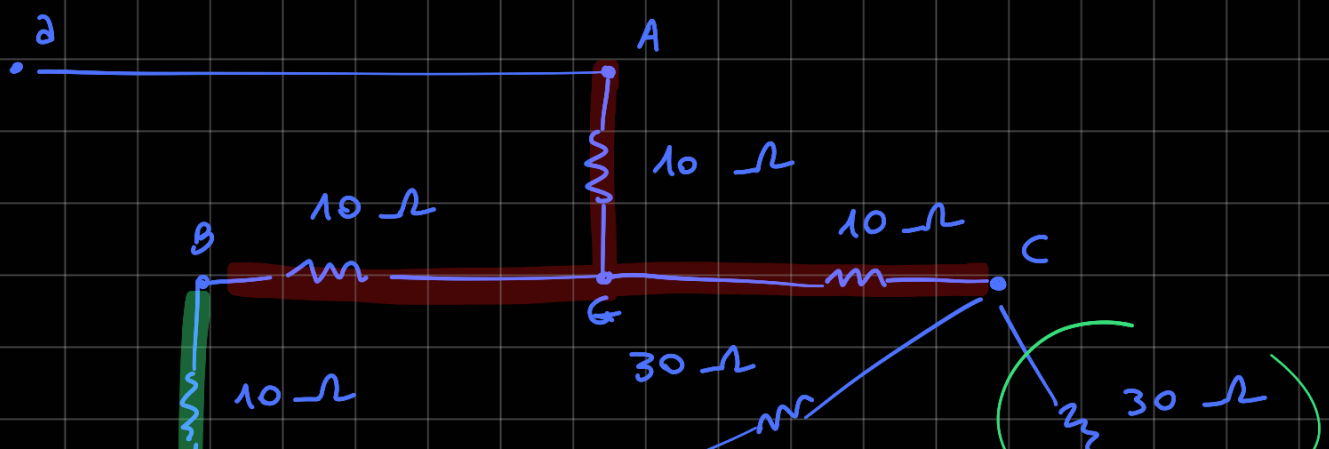
$$R_3 = \frac{R_b R_c}{R_a + R_b + R_c}$$

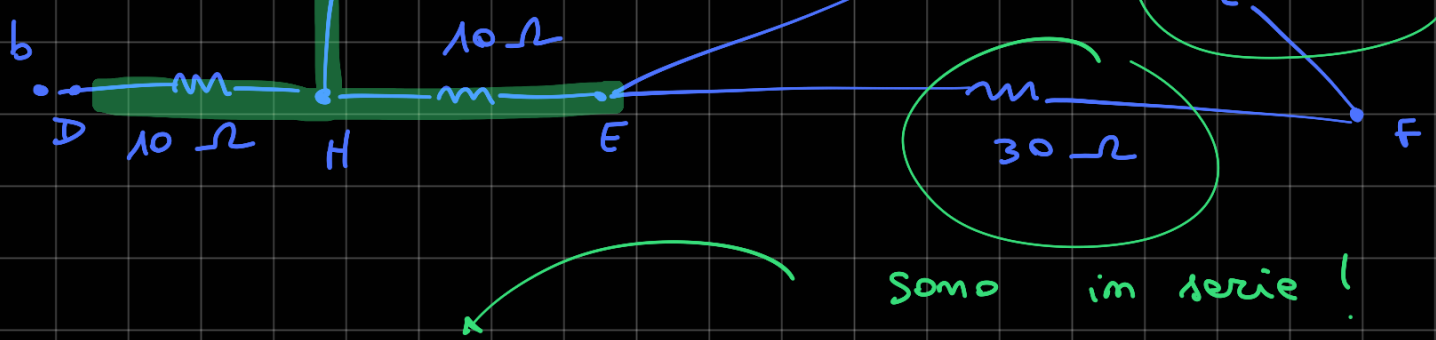
Se tutte le resistenze sono uguali, allora

$$R_{\text{STELLA}} = \frac{R_{\text{TRIANGOLO}}}{3} \Leftrightarrow R_{\text{TRIANGOLO}} = 3 R_{\text{STELLA}}$$



Trasformo due triangolini in stelle...



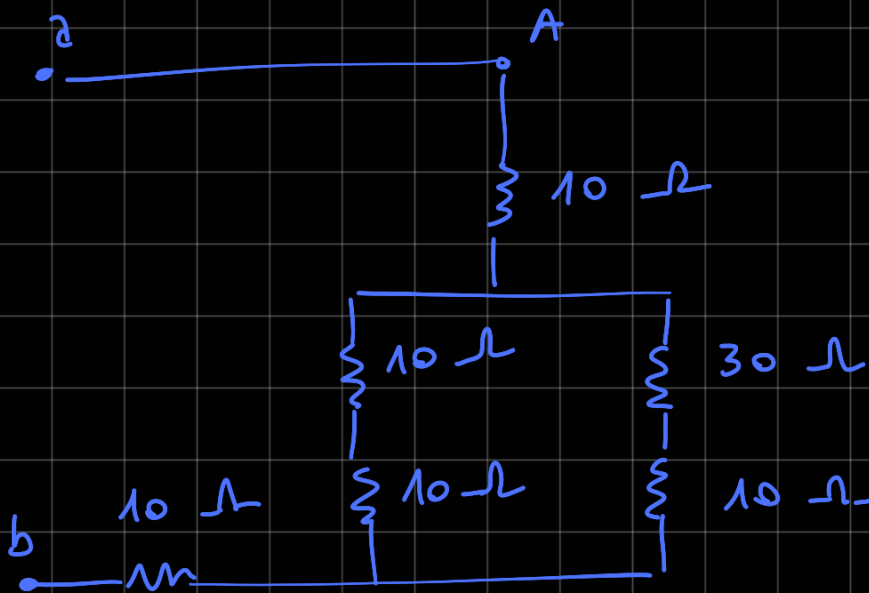


$$R_{CE} = 60 \, \Omega$$

$$\rightarrow R_{GE} = 10 + \left(\frac{1}{R_{CE}} + \frac{1}{30 \, \Omega} \right)^{-1} = 30 \, \Omega$$

Resistenza equivalente
dell'ultimo triangolo

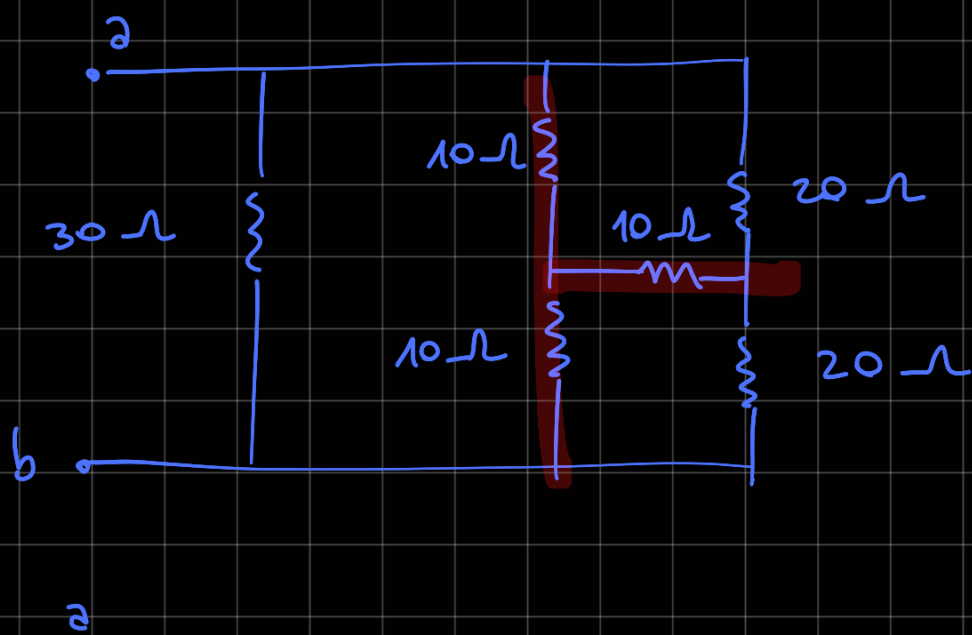
Circuito eq.:

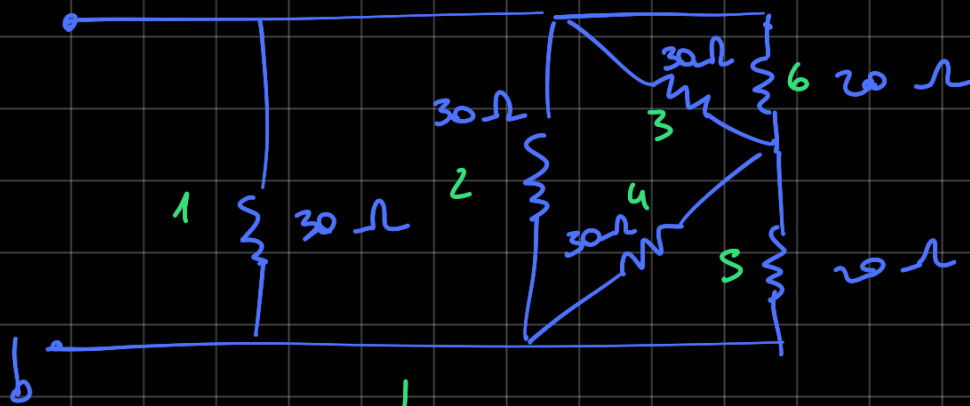


$$\Rightarrow R_{ab} = 10 + \left(\frac{1}{10+10} + \frac{1}{30+10} \right)^{-1} + 10 \approx$$

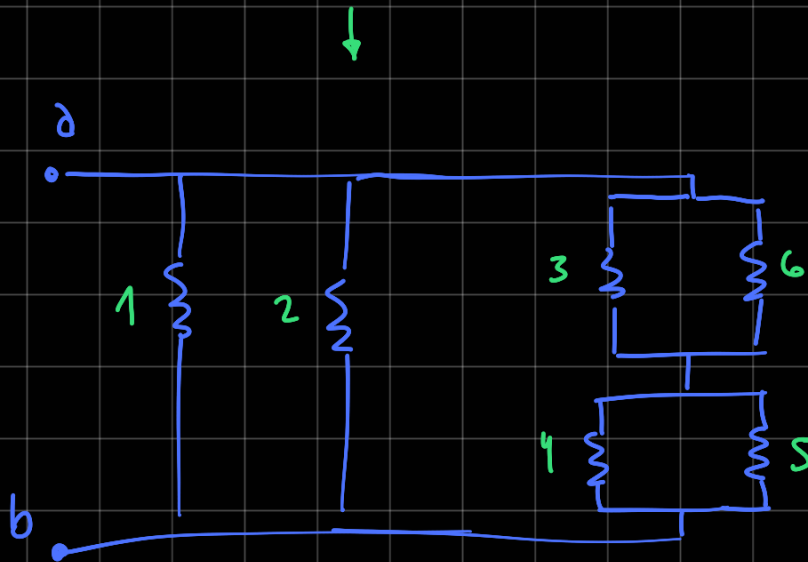
$$\approx 33,33 \, \Omega$$

12





EQUIVALENTE A...



$$R_{36} = \left(\frac{1}{R_3} + \frac{1}{R_6} \right)^{-1} = 12 \, \Omega$$

$$R_{45} = \left(\frac{1}{R_4} + \frac{1}{R_5} \right)^{-1} = 12 \, \Omega$$

$$\Rightarrow R_{eq} = \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_{36} + R_{45}} \right)^{-1} \approx 9,23 \, \Omega$$